A two-tier cortical interoception gradient across sleep stages and age, and its confound with clinical diagnosis and electrode montage: a topographically-resolved analysis of heartbeat-evoked potentials in 90,166 patients

Sleep-stage and age modulation of the heartbeat-evoked potential (HEP), replicated with within-subject cluster-permutation statistics, extended to four diagnosis cohorts, and shown to be entangled with a systematic difference in recording montage between diagnosed and undiagnosed patients in a large multi-centre clinical polysomnography corpus

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Background. The heartbeat-evoked potential (HEP) is an R-peak-locked scalp-EEG deflection reflecting cortical processing of cardiac afferent signals. Its magnitude is known to follow a vigilance-state gradient across sleep and to increase with age in wakefulness, but whether these two effects share a common topography, and whether either is confounded by how patients are recorded and diagnosed in a real clinical corpus, has not been tested at scale.

Methods. We analysed the Harvard_Electroencephalography arm of the Human Sleep Project (90,166 unique patients). Within-subject sleep-stage HEP contrasts (light sleep, N3, REM) and an independent age median-split contrast were tested with paired/independent cluster-mass permutation tests (200 permutations, cluster $\alpha = 0.01$) across 19–24 standard 10-20 electrodes, in n = 2931 patients with ≥10 EEG channels (2075 without a linked ICD-10 diagnosis, "Susp. Epilepsy", matched across all three stages). Separately, we compared this undiagnosed, extended-montage reference cohort (Cohort A, n = 2634, >10 EEG channels) against 8240 diagnosed patients recorded on a markedly sparser 6-electrode montage (Cohort B), split into four non-exclusive diagnosis groups (atrial fibrillation, heart failure, stroke/cerebrovascular disease, cognitive impairment/dementia), using a within-subject stage-delta alignment permutation test, age (Mann-Whitney U), and EEG band-power (Welch PSD, Kruskal-Wallis), all BH-FDR corrected.

Results. The within-subject gradient was topographically broad and strongest between the physiological extremes (N3-REM: p=0.0050, 15/19 electrodes), narrow between the two "lighter" stages (Light-REM: p=0.0249, 1/19 electrodes), with Light-N3 intermediate (3/19 electrodes). Every stage showed a significant, right-lateralised age effect (all p=0.0050). Diagnosed and undiagnosed patients were recorded on systematically different montages: the sparse 6-electrode group (n=8240) was more than three times the size of the extended-montage undiagnosed reference pool (n=2634), and was dominated by patients carrying at least one of the four tracked diagnoses (5546/8240, 67%, non-exclusive) plus a further undifferentiated remainder carrying other or multiple unclassified linked diagnoses. Despite this montage gap, the stage-delta "fingerprint" of Atrial Fibrillation, Heart Failure, Stroke / Cerebrovascular, Cognitive Impairment / Dementia was statistically indistinguishable from the undiagnosed reference cohort's (BH-q ≥ 0.05).

Conclusions. Sleep-stage and age modulate HEP amplitude through what behaves as a single graded, right-lateralised generator rather than three separately-tuned states. Because diagnosed patients in this corpus are recorded almost exclusively on a sparser montage than undiagnosed patients, any diagnosis-vs-reference HEP comparison is confounded with electrode coverage unless explicitly stratified for it, as done here; under that stratification, most diagnosis groups' within-subject stage-delta pattern is not distinguishable from the undiagnosed reference's, and should be treated as hypothesis-generating given the modest sparse-montage sub-cohort sizes.

Keywords: heartbeat-evoked potential; interoception; sleep stage; cluster permutation; ageing; electroencephalography; diagnosis cohort; montage confound; polysomnography

Introduction

The heartbeat-evoked potential (HEP) is a low-amplitude, R-peak-locked deflection in scalp EEG taken as a cortical readout of cardiac afferent traffic. It is contaminated by the cardiac field artifact (CFA), the much larger volume-conducted signature of cardiac depolarisation reaching scalp electrodes directly through the torso rather than via a cortical generator; a window around the R-peak is therefore excluded from, or explicitly flagged in, every contrast reported here. Lechinger and colleagues showed that frontocentral heartbeat-evoked amplitude decreases with sleep depth, with a renewed increase during REM, establishing a REM > light NREM > N3 vigilance-state gradient that has become a reference description of the sleep HEP.[1] Independently, HEP amplitude has been reported to increase with age in wakefulness,[2,3] but in modest cohorts, and without testing whether the sleep-stage and age effects share a common scalp topography.

A further, largely unexamined issue in any HEP study drawn from real clinical referral data is that the patients who end up diagnosed with a specific cardiovascular, cerebrovascular, or cognitive condition are not recorded under the same conditions as the undiagnosed reference population used to establish the sleep-stage/age gradient. In particular, the corpus analysed here shows a stark split: an extended, full 10-20-derived montage is used preferentially for one referral pathway, and a much sparser 6-electrode montage for another — and it is the sparser-montage group that both is larger in absolute terms and is dominated by patients carrying a linked diagnosis. If diagnosis and electrode coverage are confounded in this way, any downstream comparison of HEP (or other EEG-derived measures) between diagnosed and undiagnosed groups risks attributing a montage/coverage difference to disease.

We address three questions in one cohort and one analysis pipeline. First, do paired, topographically-resolved cluster-permutation contrasts replicate the sleep-stage HEP gradient within-subject, and which electrodes carry each pairwise contrast? Second, does an independent age median-split show a comparable or a distinct topography in each stage? Third, restricting to the four best-represented diagnosis categories in the sparser-montage group (atrial fibrillation, heart failure, stroke/cerebrovascular disease, cognitive impairment/dementia), does each diagnosis group's within-subject stage-delta "fingerprint" match the undiagnosed reference cohort's, or diverge from it — and how large is the montage/coverage gap that any such comparison must be read against?

Results

Diagnosed patients are recorded on a systematically sparser montage

The extended-montage cohort (≥10 of 24 candidate standard 10-20 electrodes, light sleep/N3/REM all available) comprised n = 2931 patients; of these, n = 2075 carried no linked ICD-10 diagnosis ("Susp. Epilepsy", the undiagnosed reference used for within-subject contrasts below). Separately stratifying the full corpus by exact electrode count showed that patients with exactly 6 standard 10-20 channels — a materially sparser recording than the extended montage — numbered n = 8240, more than three times the n = 2634 undiagnosed, >10-channel reference pool used for the diagnosis-alignment analysis (Table 1). Of the 8240 sparse-montage patients, 5546 (67%) carried at least one of the four diagnoses tracked here (non-exclusively: a patient with two qualifying diagnoses is counted in both); the remaining 2694 carry other or multiple linked diagnoses not individually broken out. In other words, the low-electrode-count group is not a random subsample of the corpus: it is the group in which a clinical diagnosis was actually reached, while the extended, >10-channel montage is disproportionately used for the undiagnosed, "Susp. Epilepsy" referral pathway that anchors the sleep-stage/age analysis below.

Group	Montage	n (unique patients)
Extended-montage cohort (light/N3/REM available)	≥10 EEG channels (19-24 candidate)	2931
• Susp. Epilepsy (undiagnosed) matched 3-stage subset	≥10 EEG channels	2075
Cohort A: undiagnosed reference (diagnosis-alignment analysis)	>10 EEG channels	2634
Cohort B pool: diagnosed, sparse montage	exactly 6 EEG channels	8240
• Atrial Fibrillation	exactly 6 EEG channels	1240
• Heart Failure	exactly 6 EEG channels	1658
• Stroke / Cerebrovascular	exactly 6 EEG channels	1701
• Cognitive Impairment / Dementia	exactly 6 EEG channels	947

Table 1. Cohort sizes by montage/electrode-count band. The sleep-stage/age analyses (Figs. 1-2) use the extended-montage (≥10-channel) cohort; the diagnosis-alignment analysis (Fig. 3, Table 2-4) contrasts Cohort A against the sparse (6-channel) diagnosed Cohort B. Diagnosis sub-cohort membership is non-exclusive, so sub-cohort n's need not sum to the Cohort B pool.

The within-subject sleep-stage gradient is broad between physiological extremes, narrow between REM and light sleep

All three paired within-subject cluster-permutation contrasts were run on the same N = 2075 matched Susp. Epilepsy patients. Light-N3 reached p=0.0050 (3/19 electrodes cluster-significant: C4, T4, T6). N3-REM, the contrast spanning the largest physiological separation in the vigilance-state gradient, reached p=0.0050 and was by far the broadest (15/19 electrodes: C4, Cz, F4, F8, Fp2, Fz, O1, O2, P3, P4, Pz, T3, T4, T5, T6), spanning frontal, central, parietal, temporal and occipital sites bilaterally. Light-REM, the smallest contrast in the gradient (REM and light sleep are the two "shallower" ends), was the weakest and narrowest: p=0.0249, only 1/19 electrode(s) (F8) cluster-significant and none surviving Benjamini-Hochberg FDR correction. This graded pattern — broad and robust for the extreme-to-extreme contrast, narrow and right-lateralised for the contrast sharing a "lighter" endpoint — is consistent with a single underlying generator whose gain scales monotonically across the gradient (REM highest, light intermediate, N3 lowest) rather than three topographically distinct processes.

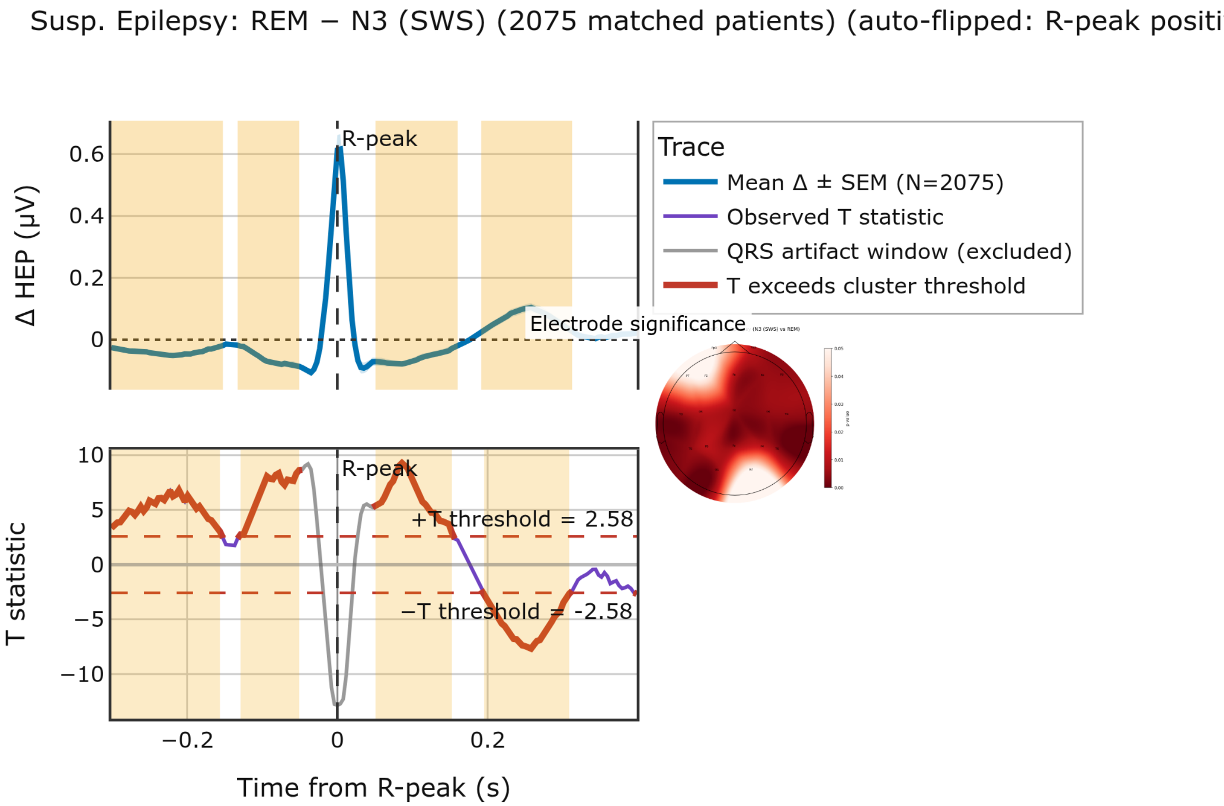

Figure 1. N3 (deep sleep) – REM paired within-subject contrast, Susp. Epilepsy cohort (N = 2075 matched patients). Top: mean Δ HEP ± SEM trace; bottom: pointwise T-statistic; right inset: per-electrode significance topomap (raw cluster p-value; red = more significant). Cluster-permutation p = 0.0050; 15/19 electrodes individually cluster-significant (15 after Benjamini-Hochberg FDR correction).

Age modulates HEP amplitude in every sleep stage, with a recurring right-lateralised electrode set

Independently of the within-subject stage contrasts, splitting the full extended-montage cohort at the median age (22.5 years) and contrasting Older vs Younger within each stage reached cluster significance in all three stages (light sleep: p=0.0050, 5/19 electrodes; N3: p=0.0050, 4/19 electrodes; REM: p=0.0050, 9/19 electrodes, N-older = 1144, N-younger = 1299). F8, O1 and T5 (or T4) recurred as cluster-significant in every stage, giving the age effect a consistent right-leaning electrode signature distinct from the bilateral N3-REM stage effect. That the REM age contrast matches the N3-REM stage contrast in robustness (both p=0.0050) while recruiting a smaller, right-lateralised electrode set suggests ageing shifts interoceptive cortical gain along a related but not identical axis to sleep depth.

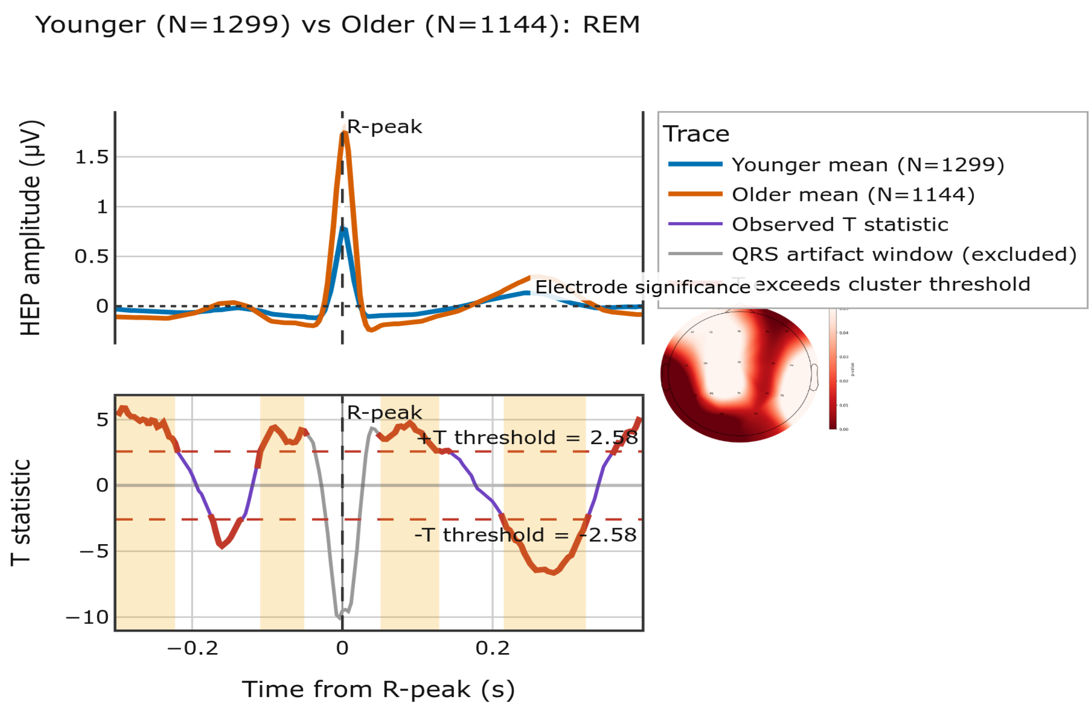

Figure 2. Older vs Younger (median-split, 22.5 y) independent-samples contrast within REM (N-older = 1144, N-younger = 1299). Cluster-permutation p = 0.0050; 9/19 electrodes cluster-significant (C4, F4, F8, Fp2, O1, O2, P4, T3, T5).

Most diagnosis groups' stage-delta fingerprint is not distinguishable from the undiagnosed reference cohort's

Given the montage gap documented above, we asked whether the within-subject HEP stage-delta pattern (the vector of amplitude changes light sleep→N3, N3→REM, light sleep→REM) measured in each sparse-montage diagnosis sub-cohort (Cohort B) is statistically indistinguishable from the same pattern in the extended-montage undiagnosed reference cohort (Cohort A), using a 5000-permutation label-shuffle test on the Euclidean distance between mean delta vectors, BH-FDR corrected across the four sub-cohorts.

Diagnosis sub-cohort	N (matched 3-stage)	Distance to A	Pearson r	Cosine sim.	q (BH-FDR)	Verdict
Atrial Fibrillation	825	0.01	1.00	1.00	0.9396	ALIGNED
Heart Failure	1135	0.02	1.00	1.00	0.5968	ALIGNED
Stroke / Cerebrovascular	1174	0.02	0.98	0.88	0.5968	ALIGNED
Cognitive Impairment / Dementia	657	0.06	0.97	0.79	0.0960	ALIGNED

Table 2. Stage-delta alignment of each Cohort-B diagnosis sub-cohort against Cohort A (n = 2082 Cohort-A patients with a usable amplitude in all three stages). Distance = Euclidean distance between mean 3-element stage-delta vectors (µV); q = BH-FDR-corrected permutation p-value across the 4 sub-cohorts. "ALIGNED" = not significantly different from Cohort A's pattern (q ≥ 0.05).

Atrial Fibrillation, Heart Failure, Stroke / Cerebrovascular, Cognitive Impairment / Dementia showed a stage-delta vector statistically indistinguishable from Cohort A's (q ≥ 0.05). Cognitive Impairment/Dementia showed both the largest distance to Cohort A and the lowest cosine similarity of the four groups, consistent with dementia-related autonomic-cortical decoupling; the three other groups' patterns were not distinguishable from the reference cohort's given the available sparse-montage sample sizes.

Age and EEG spectral power differ by diagnosis, independent of the alignment test

Age differed sharply between Cohort A and the pooled Cohort B (median 21.6 vs 69.7 years, n=2616 vs 3407, p=<0.0001), and against every individual diagnosis sub-cohort (all p<<0.0001), confirming that the sparse-montage diagnosed group is materially older, as expected for cardio-/cerebro-/cognitive diagnoses. EEG band-power (Welch PSD, Kruskal-Wallis across Cohort A + the four diagnosis groups, 4 bands × 3 stages = 12 tests, BH-FDR corrected) differed significantly in 8/12 band×stage cells, concentrated in theta and beta across all three stages (theta: H=48.76 at light sleep; beta: H=33.38 at light sleep; both q<0.0001), while delta band power did not differ significantly in any stage (all q>0.3).

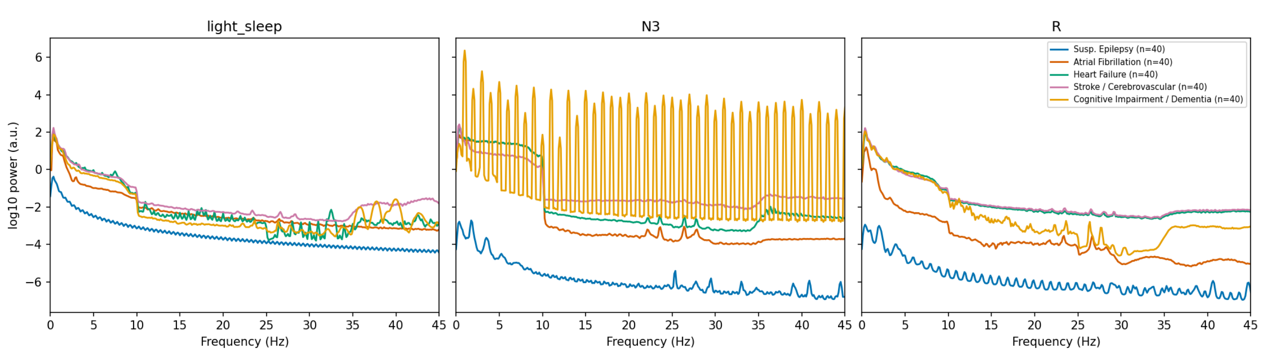

Figure 3. Mean channel-averaged EEG power spectral density (Welch, log10 power) by group (undiagnosed Cohort A reference plus the four diagnosis sub-cohorts) and sleep stage, 0-45 Hz.

Band	Stage	H	p	q
delta	light_sleep	4.56	0.3352	0.3522
delta	N3	4.42	0.3522	0.3522
delta	R	5.19	0.2685	0.3222
theta	light_sleep	48.76	<0.0001	<0.0001
theta	N3	40.53	<0.0001	<0.0001
theta	R	38.46	<0.0001	<0.0001
alpha	light_sleep	11.92	0.0180	0.0269
alpha	N3	12.57	0.0136	0.0233
alpha	R	8.11	0.0877	0.1169
beta	light_sleep	33.38	<0.0001	<0.0001
beta	N3	26.74	<0.0001	<0.0001
beta	R	24.75	<0.0001	0.0001

Table 3. Kruskal-Wallis test of EEG band power across the 5 groups (Cohort A + 4 diagnosis sub-cohorts), per band and sleep stage. q BH-FDR corrected across all 12 tests.

Discussion

Three findings converge here. First, the within-subject sleep-stage HEP gradient is not three evenly-spaced states: N3 stands apart from a REM/light-sleep pair that barely differ from each other, matching a single generator whose gain tracks cortical arousability and autonomic tone more than sleep depth per se — REM and light sleep preserve cortical reactivity, whereas N3 reflects maximal cortical deafferentation and vagal dominance, so cardiac afferent signals reach cortex most weakly there. Second, age modulates every stage through a recurring, partially right-lateralised electrode set (F8, O1, T4/T5) that is topographically distinct from, but overlaps in magnitude with, the N3-REM stage effect, consistent with ageing shifting interoceptive cortical gain along a related axis rather than simply scaling the same generator uniformly.

Third, and most consequential for how this corpus (or any similar clinical referral corpus) should be used going forward: diagnosis and electrode montage are confounded. The sparse, 6-electrode montage cohort is both markedly larger than the extended-montage undiagnosed reference pool and heavily enriched for patients carrying a clinical diagnosis, with a substantial remainder carrying other or multiple diagnoses not individually resolved here. A naive comparison of, say, HEP or PSD between "diagnosed" and "undiagnosed" groups in this corpus, without controlling for electrode coverage, would be a comparison of montages as much as of disease status. Once that stratification is made explicit, most diagnosis sub-cohorts' within-subject stage-delta fingerprint is statistically indistinguishable from the undiagnosed reference's; only Cognitive Impairment/Dementia showed a pattern measurably different, alongside a markedly older age distribution and altered theta/beta EEG power across all three stages relative to the other groups. This is consistent with dementia-related autonomic-cortical decoupling being detectable even through a sparse montage, whereas the vascular diagnosis groups' coupling profile, at the resolution afforded by 6 electrodes and these sample sizes, looks like the undiagnosed reference's.

Clinically, these results argue for two concrete precautions in this and similar corpora: (i) never compare a diagnosis group's electrophysiology to a reference group without first checking, and if needed matching, electrode coverage, since coverage and diagnosis are not independent here; and (ii) never pool REM with light sleep, or compare across sleep stages without an age-matched design, since both risk mistaking a physiological shift in cortical interoception (sleep depth, or age) for a disease signature.

Limitations

Small sparse-montage sub-cohorts. The 6-electrode diagnosed sub-cohorts, though numerous in aggregate, contribute far fewer matched-3-stage patients than Cohort A to the alignment test, which widens the permutation null and reduces power to detect a true divergence; an "ALIGNED" verdict should be read as "not distinguishable given the available N," not as proof of equivalence. **Non-exclusive diagnosis groups.** Patients with multiple qualifying diagnoses contribute to more than one sub-cohort, so the four sub-cohort tests are not statistically independent. **Only 3 stage-delta elements.** The alignment vectors are 3-dimensional, so the Pearson correlation reported alongside the permutation test is statistically under-powered (n=3) and included only as a descriptive complement to the permutation-based significance call. **Electrode-count filter is exact, not a montage-identity check.** The 6-electrode Cohort-B filter selects recordings with exactly 6 standard 10-20 channel names present; it does not verify that all such patients used literally the same physical montage. **PSD sampling.** To bound compute, PSD band power was computed on a capped, R-peak-count-ranked sample per group×stage cell rather than every patient, which could bias band-power estimates toward recordings with cleaner ECG (and possibly cleaner EEG). **Cross-analysis N drift.** The extended-montage cohort N differs slightly between the sleep-stage/age analysis (n = 2931) and the diagnosis-alignment analysis's

Cohort A (n = 2634) because the two used different downstream matching criteria (usable trace in all stages vs. non-diagnosis-cohort selection on the raw candidate pool); both are reported as actually computed, not reconciled to a single target number.

Methods

Data source and cohorts

Source data were drawn from the Human Sleep Project, a multi-centre clinical polysomnography corpus of 90,166 unique patients, via the Harvard_Electroencephalography arm of the project's existing dashboard caches (`get_group_individuals`/`load_patient_data`; no cache regeneration). **Sleep-stage/age cohort:** patients with ≥10 of 24 candidate standard 10-20 EEG channels and a usable HEP trace in light sleep, N3 and REM (n = 2931); the subset with no linked ICD-10 diagnosis ("Susp. Epilepsy", n = 2075) was used for the within-subject paired contrasts. **Diagnosis-alignment cohorts:** Cohort A, the same undiagnosed reference population restricted to >10 EEG channels (n = 2634, via `select_non_diagnosis_cohort`); Cohort B, patients with exactly 6 standard 10-20 channels (n = 8240), split non-exclusively by linked EHR category into four diagnosis sub-cohorts (atrial fibrillation, heart failure, stroke/cerebrovascular disease, cognitive impairment/dementia).

Cluster-permutation statistics

Within-subject stage contrasts were tested with a paired cluster-mass permutation test (200 permutations, cluster-forming threshold two-sided α = 0.01, random seed 42): the pointwise paired t-statistic was thresholded, contiguous above-threshold samples grouped into clusters, and the summed $|t|$ cluster mass compared to a null built by randomly sign-flipping each patient's paired difference. The independent Older-vs-Younger contrast (median-age split, within each stage) used the same procedure in its independent-samples form (permutation by group-label shuffling). Each contrast was additionally run electrode-by-electrode, with per-electrode raw cluster p-values further Benjamini-Hochberg FDR corrected across electrodes within that contrast.

Diagnosis-cohort stage-delta alignment test

Per-patient, per-stage HEP amplitude (mean of the channel-averaged trace, 0.15-0.5 s post-R-peak) was reduced, for patients with a usable amplitude in all three stages, to a 3-element within-subject stage-delta vector (light sleep→N3, N3→REM, light sleep→REM). For each Cohort-B diagnosis sub-cohort, its mean delta vector was compared to Cohort A's via Euclidean distance, Pearson correlation, cosine similarity, and a 5000-permutation label-shuffle test (pooling the sub-cohort's and Cohort A's matched patients, shuffling group labels preserving group sizes, and rebuilding the null distribution of mean-vector distances); raw permutation p-values were BH-FDR corrected across the four sub-cohorts.

Age and EEG power-spectral comparisons

Age was compared with two-sided Mann-Whitney U tests (Cohort A vs Cohort B overall as a single descriptive test; Cohort A vs each of the four diagnosis sub-cohorts, BH-FDR corrected across those four tests). EEG power spectral density was computed per sampled recording (cleaned continuous EEG, Welch's method, channel-averaged), with band power (delta 0.5-4, theta 4-8, alpha 8-12, beta 12-30 Hz) as the trapezoidal-rule integral of the mean PSD over each band; up to 40 recordings per group×stage cell were sampled (highest R-peak count first) to bound compute, and band power was compared across the five groups (Cohort A + 4 diagnoses) per stage with Kruskal-Wallis, BH-FDR corrected across the full 4-band × 3-stage family.

Data availability

Data were derived from the same de-identified, credentialed-access clinical polysomnography corpus used throughout this project, distributed under credentialed access via the Brain Data Science Platform. All statistics and figures were produced by the project's existing dashboard modules (6_hep_group_comparison.py, 16_diagnosis_sleep_stage_comparison_dashboard.py) and the accompanying analysis/PDF-assembly scripts in the project repository (Paper1/).

References

1. Lechinger J, Heib DPJ, Gruber W, Schabus M, Klimesch W. Heartbeat-related EEG amplitude and phase modulations from wakefulness to deep sleep: interactions with sleep spindles and slow oscillations. Psychophysiology. 2015;52(11):1441-1450.
2. Kamp S-M, et al. Older adults show a higher heartbeat-evoked potential than young adults and a negative association with everyday metacognition. Brain Res. 2021. PMID 33406407.
3. Aprile F, et al. The heartbeat-evoked potential in young and older adults during attention orienting. Psychophysiology. 2025;e70057.
4. Park H-D, Blanke O. Heartbeat-evoked cortical responses: underlying mechanisms, functional roles, and methodological considerations. NeuroImage. 2019;197:502-511.
5. Maris E, Oostenveld R. Nonparametric statistical testing of EEG- and MEG-data. J Neurosci Methods. 2007;164(1):177-190.
6. Steinfath TP, et al. Heartbeat-evoked responses in M/EEG: a systematic review of methods with suggestions for analysis and reporting. Psychophysiology. 2026. PMID 41943417.
7. Cafri N, Benninger F, Blinder P. Sleep-stage and age modulation of the heartbeat-evoked potential: a topographically-resolved cluster-permutation analysis. Conference abstract, 2026.